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09: Continuous RVs

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January 29, 2024

[Lecture Discussion on Ed](#)



Continuous RVs

People heights

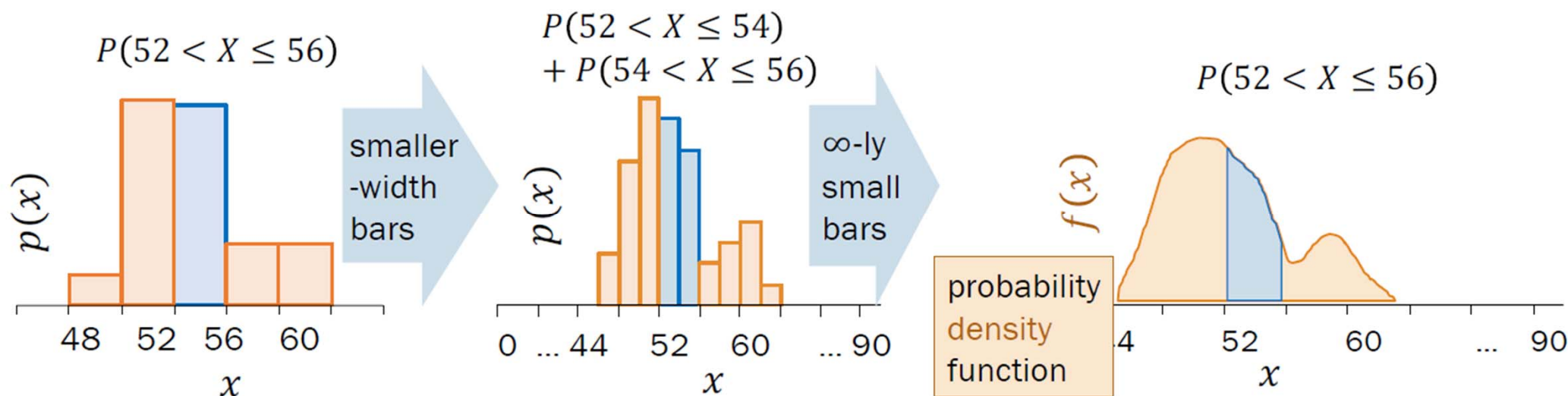
You are volunteering at the local elementary school.

- To buy a t-shirt for your friend Chendi, you need to know her height.

1. What is the probability that your friend is 54.0923857234 inches tall?

Essentially 0

2. What is the probability Chendi is between 52–56 inches tall?



Continuous RV definition

A random variable X is **continuous** if there is a **probability density function** $f(x) \geq 0$ such that for $-\infty < x < \infty$:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Integrating a PDF must always yield a valid probability, no matter the values of a and b . The PDF must also satisfy:

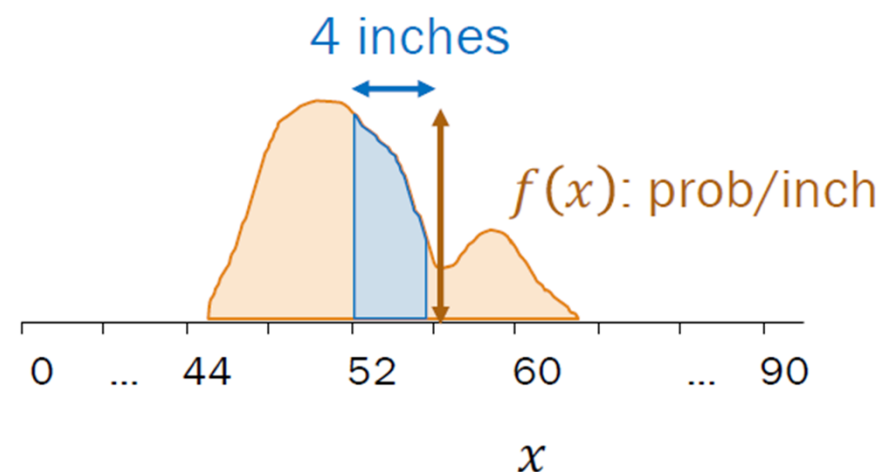
$$\int_{-\infty}^{\infty} f(x) dx = P(-\infty < X < \infty) = 1$$

Note: $f(x)$ is sometimes written as $f_X(x)$ to be clear the random variable is X .

Main takeaway #1

Integrate $f(x)$ to get probabilities.

PDF Units: probability per units of X



$$P(52 \leq X \leq 56) = \int_{52}^{56} f(x) dx$$

PMF vs PDF

Discrete random variable X

Probability mass function (PMF):

$$p(x)$$

To get probability:

$$P(X = x) = p(x)$$

Continuous random variable X

Probability density function (PDF):

$$f(x)$$

To get probability:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Both are measures of how **likely** X is to take on a value or some range of values.

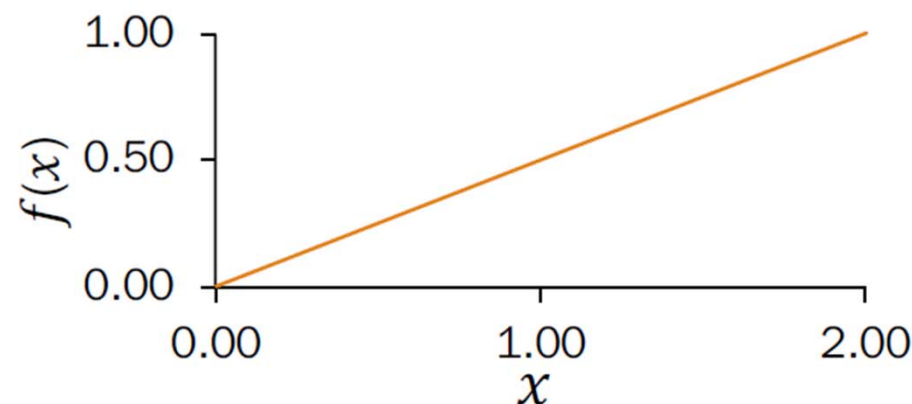
Computing probability

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Let X be a continuous RV with PDF:

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

What is $P(X \geq 1)$?



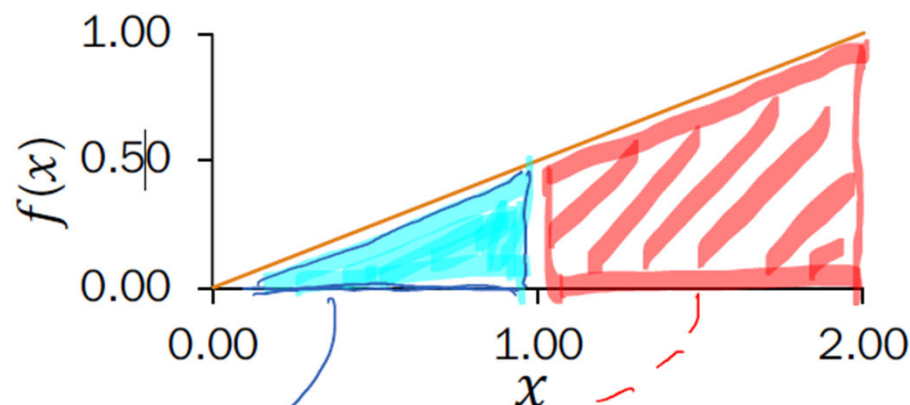
Computing probability

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Let X be a continuous RV with PDF:

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

confirm: $\int_0^2 \frac{x}{2} dx = 1$, so valid pdf



What is $P(X \geq 1)$?

Strategy 1: Integrate

$$P(1 \leq X < \infty) = \int_1^{\infty} f(x) dx = \int_1^2 \frac{1}{2} x dx$$
$$= \frac{1}{2} \left(\frac{1}{2} x^2 \right) \Big|_1^2 = \frac{1}{2} \left[2 - \frac{1}{2} \right] = \frac{3}{4}$$

Strategy 2: Know triangles

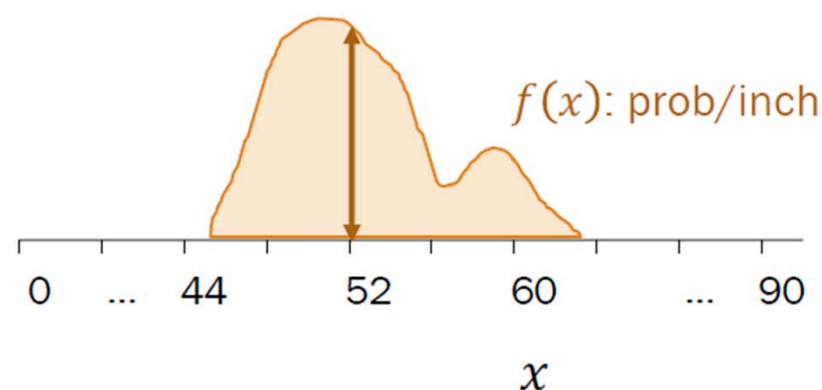
$$1 - \frac{1}{2} \left(\frac{1}{2} \right) = \frac{3}{4}$$

Wait! Is this even legal?

$$P(0 \leq X < 1) = \int_0^1 f(x) dx ??$$

Main takeaway #2

For a continuous random variable X with PDF $f(x)$,
$$P(X = c) = \int_c^c f(x)dx = 0.$$

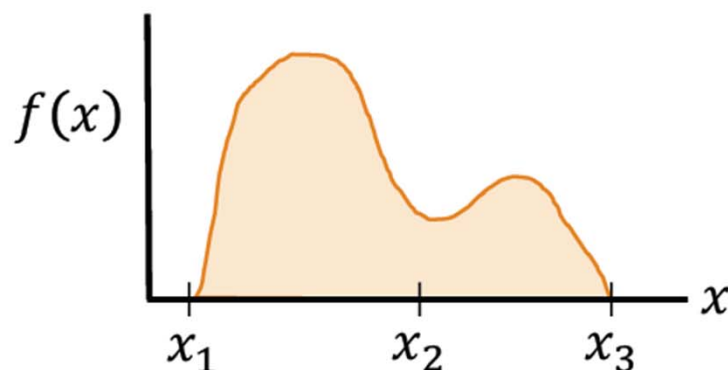


Contrast with PMF in discrete case: $P(X = c) = p(c)$

PDF Properties

For a **continuous** RV X with PDF f ,

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$



support: values of x where $f(x) > 0$

Interval width $dx \rightarrow 0$

True/False:

- ★ 1. $P(X = c) = 0$
- ★ 2. $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b) = P(a < X \leq b)$
- ✗ 3. $f(x)$ is a probability
- ★ 4. In the graphed PDF above,
 $P(x_1 \leq X \leq x_2) > P(x_2 \leq X \leq x_3)$

It's a probability density!

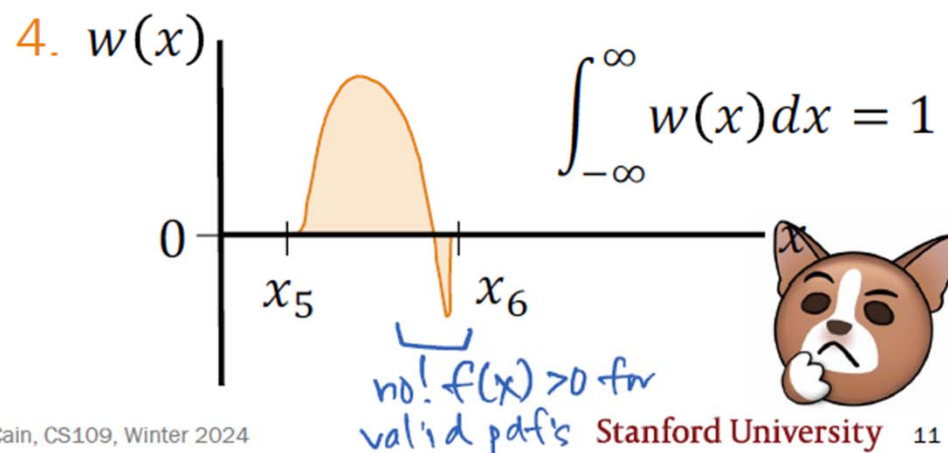
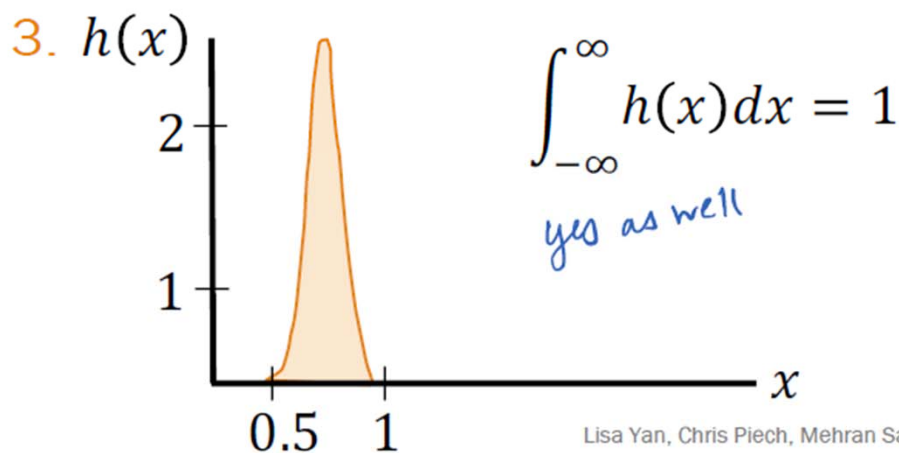
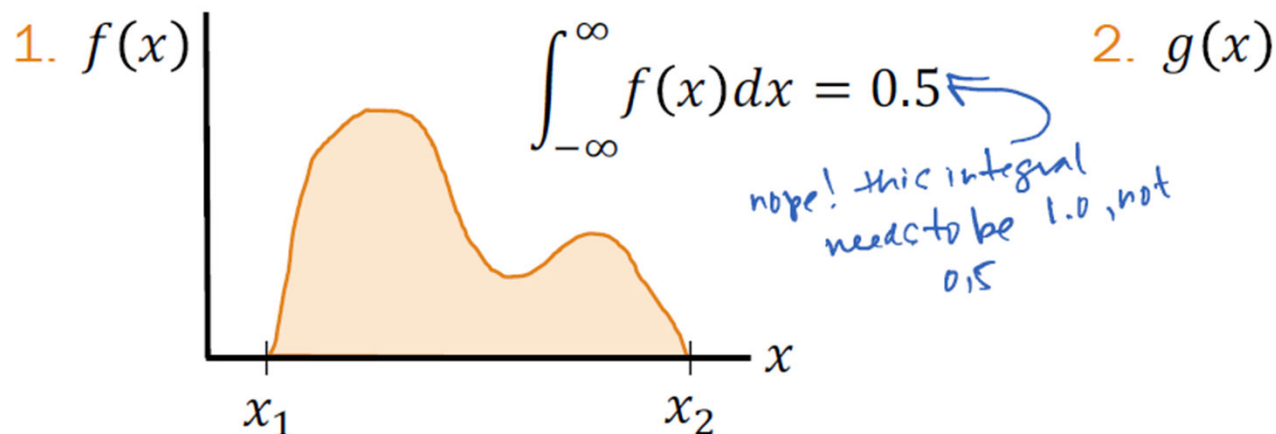
Compare area under the curve



Determining valid PDFs

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Which of the following functions are valid PDFs?





Uniform RV

Uniform Random Variable

def A **Uniform** random variable X is defined as follows:

$$X \sim \text{Uni}(\alpha, \beta)$$

PDF

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

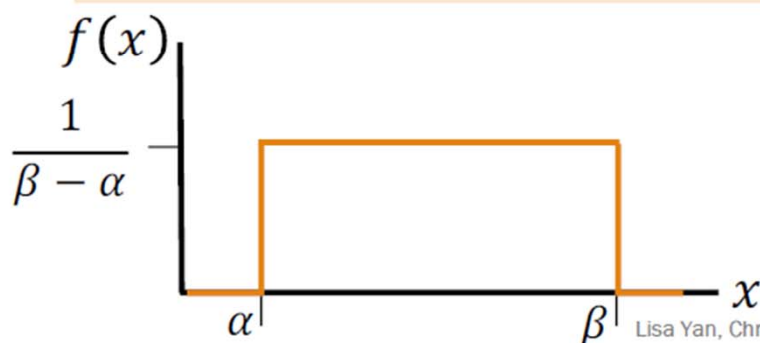
Support: $[\alpha, \beta]$
(sometimes defined
over (α, β))

Expectation

$$E[X] = \frac{\alpha + \beta}{2}$$

Variance

$$\text{Var}(X) = \frac{(\beta - \alpha)^2}{12}$$

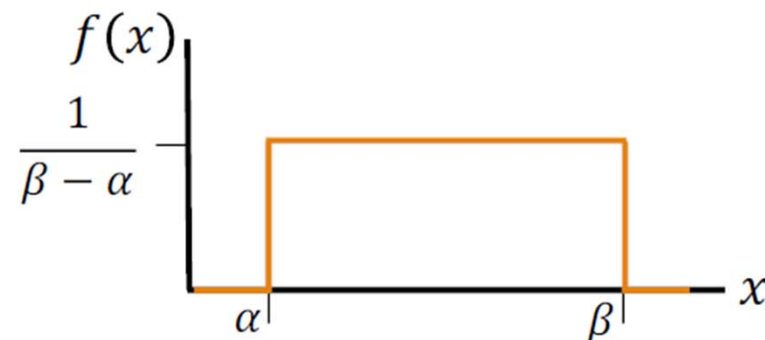


Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Winter 2023

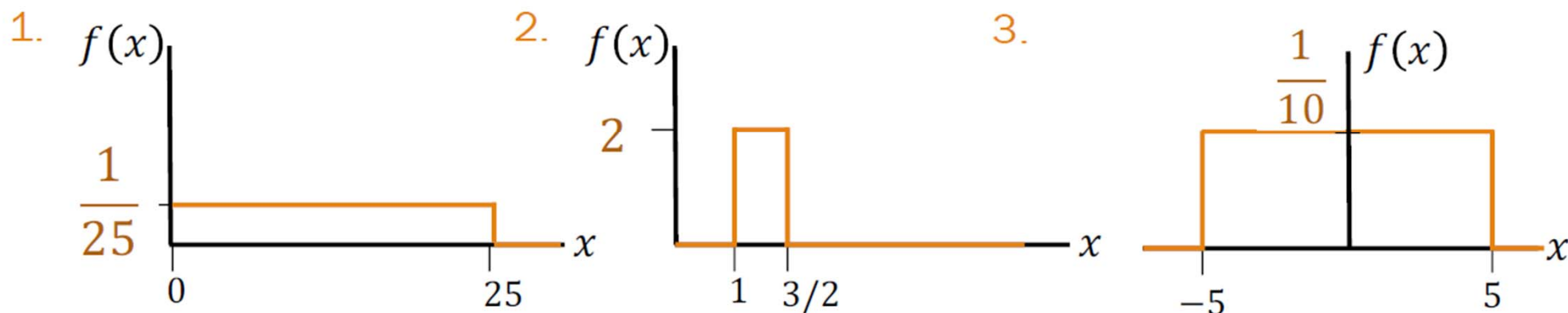
Quick check

If $X \sim \text{Uni}(\alpha, \beta)$, the PDF of X is:

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$



What is $\frac{1}{\beta - \alpha}$ if the following graphs are PDFs of Uniform RVs X ?



Expectation and Variance

Discrete RV X

$$E[X] = \sum_x x p(x)$$

$$E[g(X)] = \sum_x g(x) p(x)$$

Continuous RV X

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Both continuous and discrete RVs

$$E[aX + b] = aE[X] + b$$

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

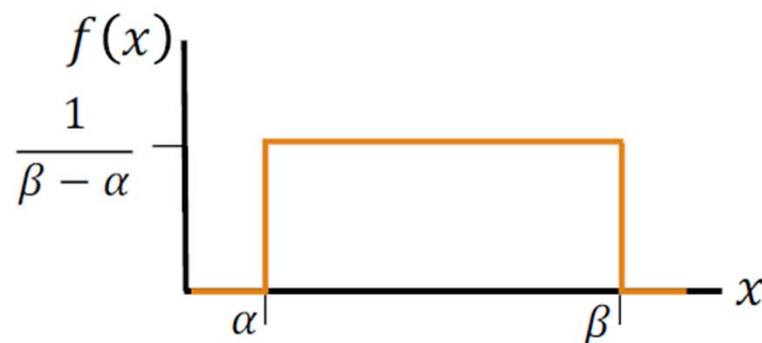
$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

} Linearity of
Expectation
} Properties of
variance

$$\text{TL;DR: } \sum_{x=a}^b \Rightarrow \int_a^b$$

Uniform RV expectation

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x \cdot f(x) dx \\ &= \int_{\alpha}^{\beta} x \cdot \frac{1}{\beta - \alpha} dx \\ &= \frac{1}{\beta - \alpha} \cdot \frac{1}{2} x^2 \Big|_{\alpha}^{\beta} \\ &= \frac{1}{\beta - \alpha} \cdot \frac{1}{2} (\beta^2 - \alpha^2) \\ &= \frac{1}{2} \cdot \frac{(\beta + \alpha)(\beta - \alpha)}{\beta - \alpha} = \frac{\alpha + \beta}{2} \end{aligned}$$



Interpretation:
Average the start & end

Uniform Random Variable

def An Uniform random variable X is defined as follows:

$$X \sim \text{Uni}(\alpha, \beta)$$

PDF

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

Support: $[\alpha, \beta]$
(sometimes defined
over (α, β))

Expectation

$$E[X] = \frac{\alpha + \beta}{2}$$



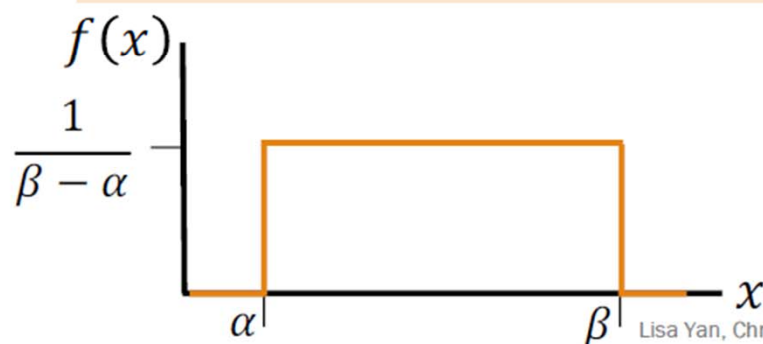
Just now

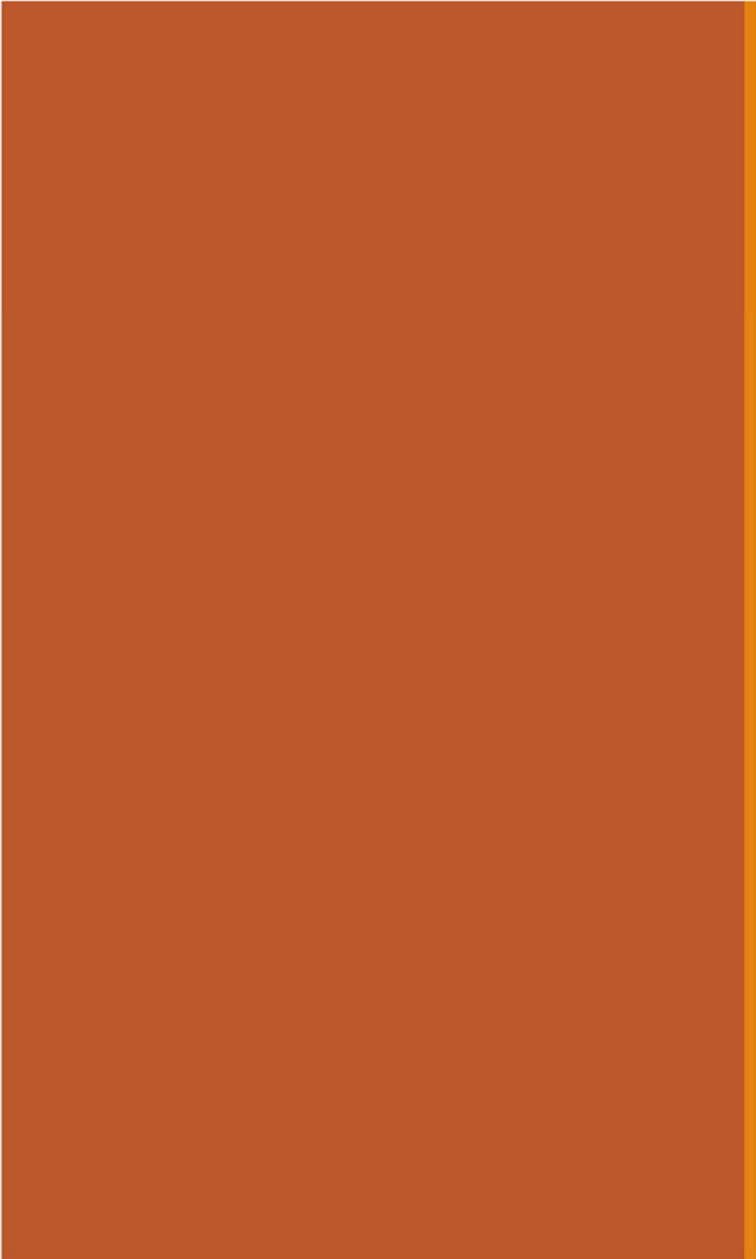
Variance

$$\text{Var}(X) = \frac{(\beta - \alpha)^2}{12}$$



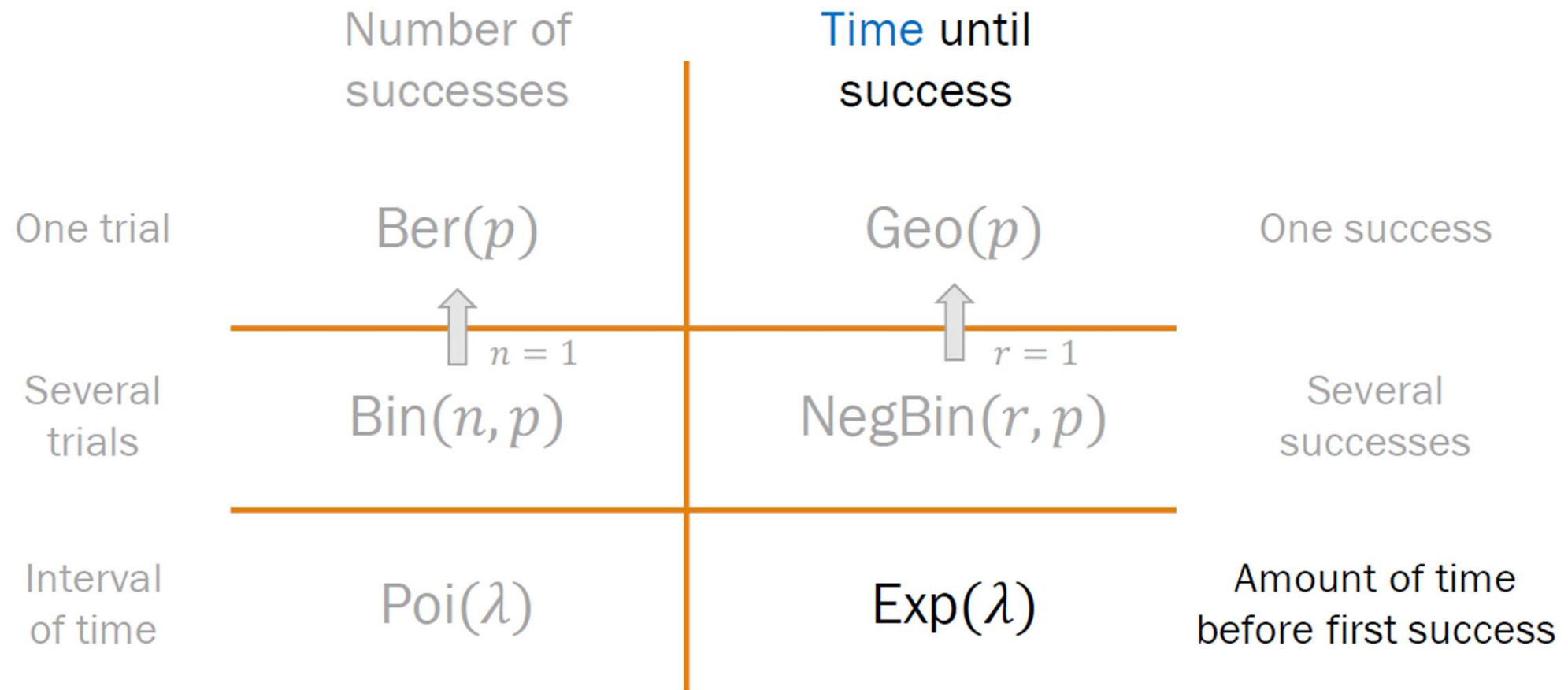
On your own time





Exponential RV

Grid of random variables



Exponential Random Variable

Consider an experiment that lasts a duration of time until success occurs.

def An **Exponential** random variable X is the amount of time until success.

$$X \sim \text{Exp}(\lambda)$$

Support: $[0, \infty)$

PDF

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Expectation

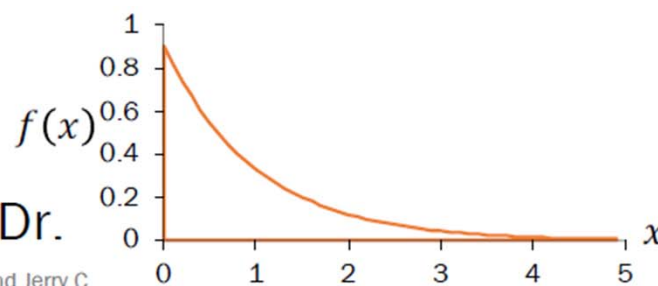
$$E[X] = \frac{1}{\lambda} \quad (\text{in extra slides})$$

Variance

$$\text{Var}(X) = \frac{1}{\lambda^2} \quad (\text{on your own})$$

Examples:

- Time until next earthquake
- Time for request to reach web server
- Time until water main break on Campus Dr.



Interpreting $\text{Exp}(\lambda)$

def An **Exponential** random variable X is the amount of time until success.

$$X \sim \text{Exp}(\lambda) \quad \text{Expectation} \quad E[X] = \frac{1}{\lambda}$$

Based on the expectation $E[X]$, what are the units of λ ?

e.g., average # of successes per second



For both Poisson and Exponential RVs,
 $\lambda = \# \text{ successes/time}$.

Earthquakes



ILL. No. 65. MEMORIAL ARCH, WITH CHURCH IN BACKGROUND, STANFORD UNIVERSITY, SHOWING TYPES OF CARVED WORK WITH THE SANDSTONE.

1906 Earthquake
Magnitude 7.8

Earthquakes

$$X \sim \text{Exp}(\lambda) \quad \begin{aligned} E[X] &= 1/\lambda \\ f(x) &= \lambda e^{-\lambda x} \quad \text{if } x \geq 0 \end{aligned}$$

on average

Major earthquakes (magnitude 8.0+) occur^{on average} once every 500 years.*

1. What is the probability of a major earthquake in the next 30 years?

We know on average:

$$\begin{aligned} 500 & \frac{\text{years}}{\text{earthquake}} \\ 0.002 & \frac{\text{earthquakes}}{\text{year}} \\ 1 & \frac{\text{earthquakes}}{500 \text{ years}} \end{aligned}$$

} if earthquakes are "successes",
then $\frac{1}{500} = 0.002$ is our lambda value,
since λ is the average number of successes
per time unit (in this case, a year).

= 1 $\frac{\text{"success"}}{500 \text{ years}}$

*In California, according to historical data from USGS, 2015

Earthquakes

$$X \sim \text{Exp}(\lambda) \quad \begin{aligned} E[X] &= 1/\lambda \\ f(x) &= \lambda e^{-\lambda x} \quad \text{if } x \geq 0 \end{aligned}$$

Major earthquakes (magnitude 8.0+) occur once every 500 years.*

1. What is the probability of a major earthquake in the next 30 years?

Define events/
RVs & state goal

X : when next
earthquake happens

$X \sim \text{Exp}(\lambda = 0.002)$

λ : $\text{year}^{-1} = 1/500$

Want: $P(X < 30)$

Solve $P(X < 30) = \int_0^{30} 0.002 e^{-0.002x} dx$

$$= 0.002 \frac{-1}{0.002} e^{-0.002x} \Big|_0^{30}$$
$$= - (e^{-0.06} - e^{0.00})$$
$$= 1 - e^{-0.06} \approx 0.058$$

Recall

$$\int e^{cx} dx = \frac{1}{c} e^{cx}$$

*In California, according to historical data from USGS, 2015

Earthquakes

$$X \sim \text{Exp}(\lambda) \quad \begin{aligned} E[X] &= 1/\lambda \\ f(x) &= \lambda e^{-\lambda x} \quad \text{if } x \geq 0 \end{aligned}$$

Major earthquakes (magnitude 8.0+) occur once every 500 years.*

1. What is the probability of a major earthquake in the next 30 years?
2. What is the **standard deviation** of years until the next earthquake?

Define events/
RVs & state goal

X : when next
earthquake happens

$X \sim \text{Exp}(\lambda = 0.002)$

λ : year⁻¹

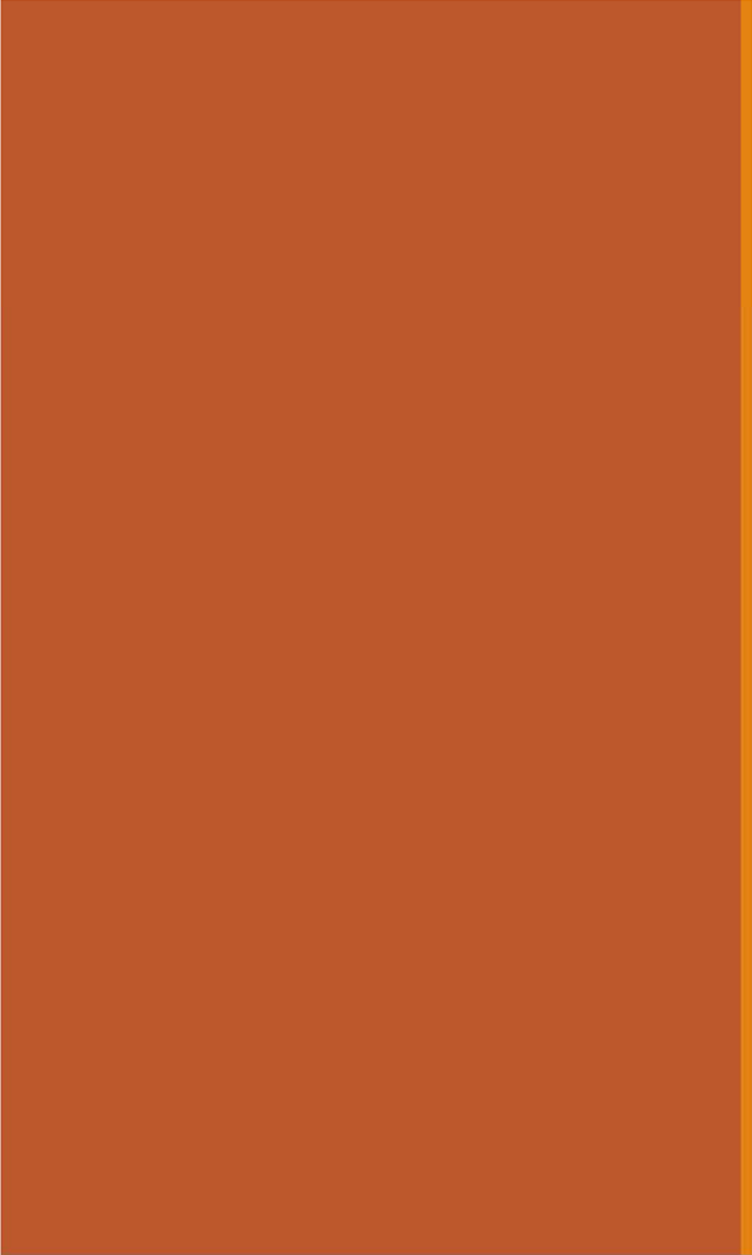
Want: $P(X < 30)$

Solve $\text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{(0.002 \text{ year}^{-1})^2} = 250,000 \text{ years}^2$

$$\text{SD}(X) = \sqrt{\text{Var}(X)} = \boxed{500 \text{ years}}$$

In general, $\text{SD}(X) = E[X] = \frac{1}{\lambda}$
whenever $X \sim \text{Exp}(\lambda)$

*In California, according to historical data from USGS, 2015

A solid orange horizontal bar spanning the width of the slide, positioned at the top.

Cumulative Distribution Functions

Cumulative Distribution Function (CDF)

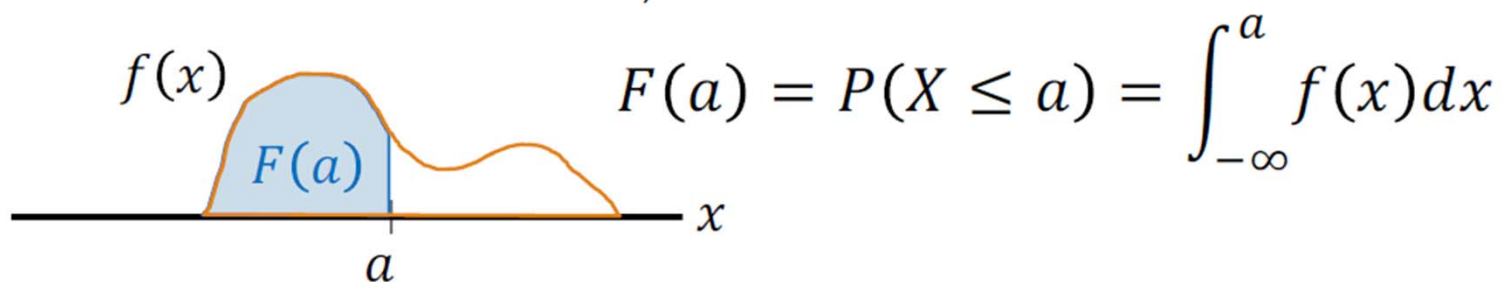
For a random variable X , the cumulative distribution function (CDF) is defined as

$$F(a) = F_X(a) = P(X \leq a), \text{ where } -\infty < a < \infty$$

For a discrete RV X , the CDF is:

$$F(a) = P(X \leq a) = \sum_{\text{all } x \leq a} p(x)$$

For a continuous RV X , the CDF is:



CDF is a probability,
though PDF is not.

If you learn to use
CDFs, you can avoid
integrating the PDF.

Using the CDF for continuous RVs

For a continuous random variable X with PDF $f(x)$, the CDF of X is

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x) dx$$

Matching (choices are used 0/1/2 times)

- | | |
|-------------------------|------------------|
| 1. $P(X < a)$ | A. $F(a)$ |
| 2. $P(X > a)$ | B. $1 - F(a)$ |
| 3. $P(X \geq a)$ | C. $F(b) - F(a)$ |
| 4. $P(a \leq X \leq b)$ | D. $F(a) - F(b)$ |



Using the CDF for continuous RVs

For a continuous random variable X with PDF $f(x)$, the CDF of X is

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x)dx$$

Matching (choices are used 0/1/2 times)

- | | | | |
|-------------------------|-------|------------------|--------------|
| 1. $P(X < a)$ | ————— | A. $F(a)$ | |
| 2. $P(X > a)$ | ————— | B. $1 - F(a)$ | |
| 3. $P(X \geq a)$ | ————— | C. $F(b) - F(a)$ | (next slide) |
| 4. $P(a \leq X \leq b)$ | ————— | D. $F(a) - F(b)$ | |

Using the CDF for continuous RVs

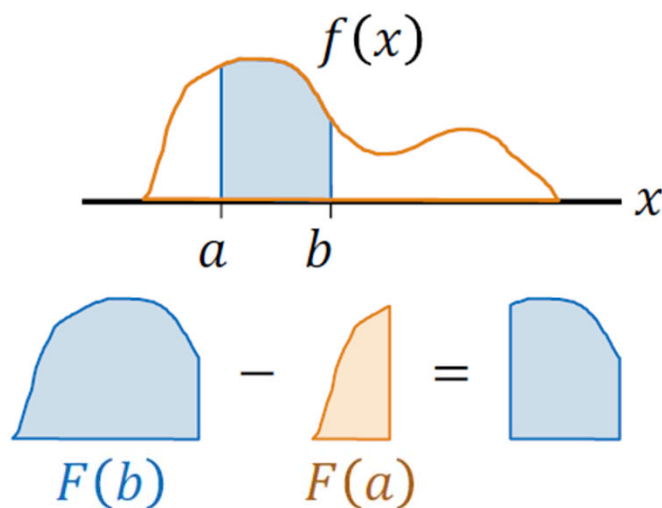
For a continuous random variable X with PDF $f(x)$, the CDF of X is

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x)dx$$

4. $P(a \leq X \leq b) = F(b) - F(a)$

Proof:

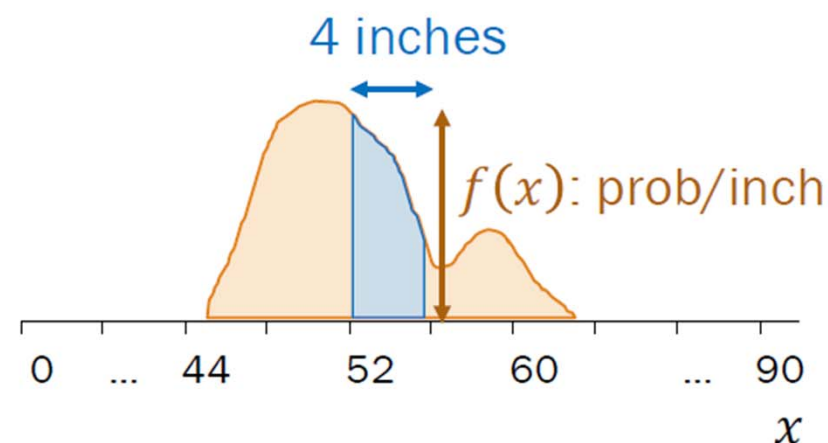
$$\begin{aligned} F(b) - F(a) &= \int_{-\infty}^b f(x)dx - \int_{-\infty}^a f(x)dx \\ &= \left(\int_{-\infty}^a f(x)dx + \int_a^b f(x)dx \right) - \int_{-\infty}^a f(x)dx \\ &= \int_a^b f(x)dx \end{aligned}$$



Addendum to main takeaway #1

Integrate $f(x)$ to get probabilities.*

*If you have $F(a)$, you already have probabilities, since $F(a) = \int_{-\infty}^a f(x) dx$



$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

CDF of an Exponential RV

$$X \sim \text{Exp}(\lambda) \quad f(x) = \lambda e^{-\lambda x} \quad \text{if } x \geq 0$$

$$X \sim \text{Exp}(\lambda) \quad F(x) = 1 - e^{-\lambda x} \quad \text{if } x \geq 0$$

Proof:

remember that $f(y) = 0$ for $y < 0$

Recall

$$\int e^{cx} dx = \frac{1}{c} e^{cx}$$

$$F(x) = P(X \leq x) = \int_{y=-\infty}^x f(y) dy = \int_{y=0}^x \lambda e^{-\lambda y} dy$$

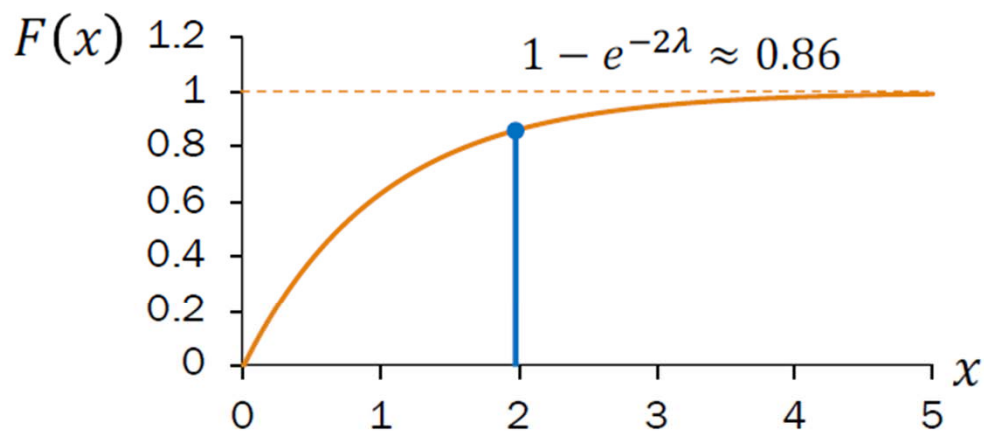
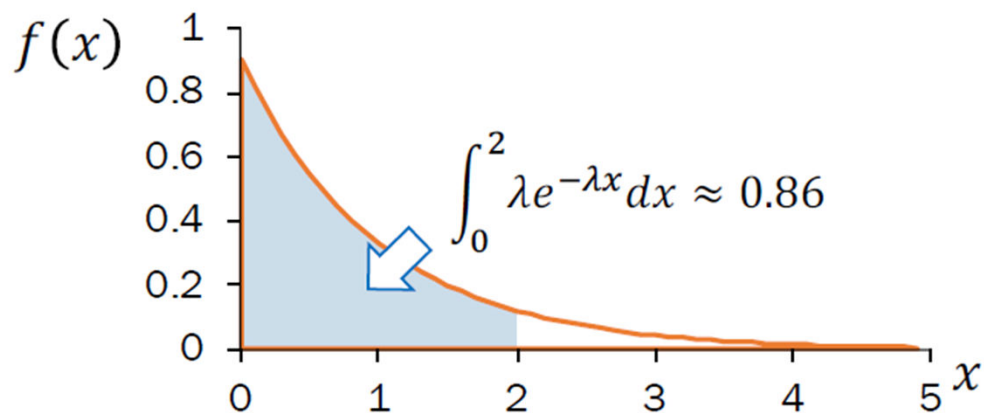
$$= \lambda \frac{1}{-\lambda} e^{-\lambda y} \Big|_0^x$$

$$= -1(e^{-\lambda x} - e^{-\lambda 0})$$

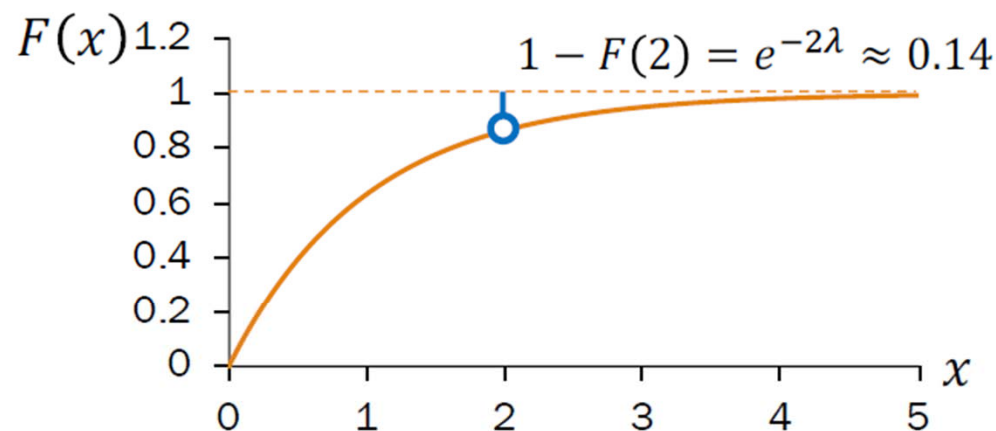
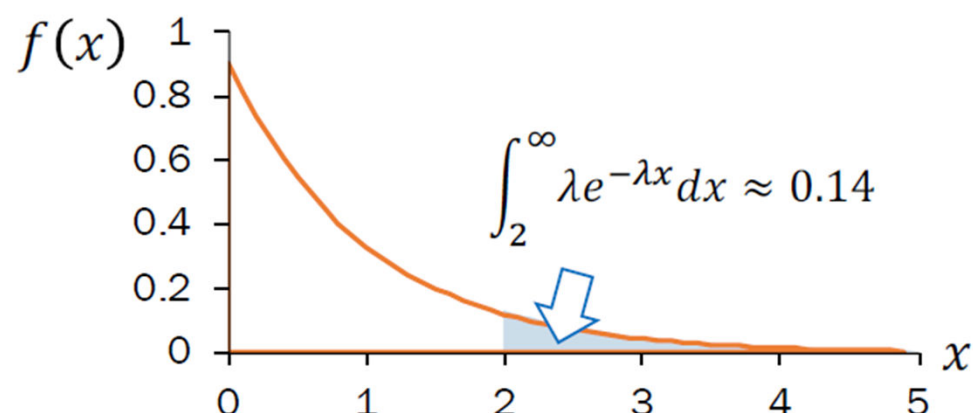
$$= 1 - e^{-\lambda x}$$

PDF/CDF $X \sim \text{Exp}(\lambda = 1)$

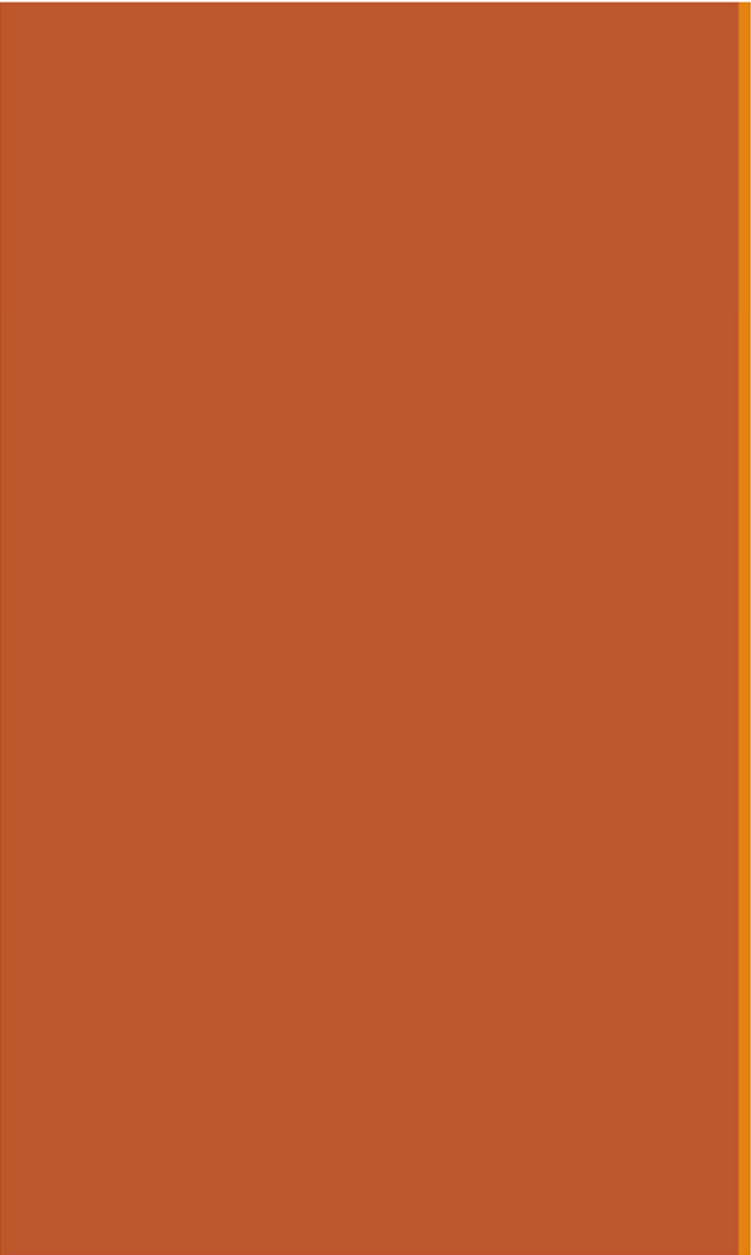
$$X \sim \text{Exp}(\lambda) \quad \begin{array}{l} x \geq 0: f(x) = \lambda e^{-\lambda x} \\ F(x) = 1 - e^{-\lambda x} \end{array}$$



$$P(X \leq 2)$$



$$P(X > 2)$$



Memoryless Property

Memorylessness: Hurry Up and Wait

A continuous probability distribution is said to be **memoryless** if a random variable X on that probability distribution satisfies the following for all $s, t \geq 0$:

$$P(X \geq s + t \mid X \geq s) = P(X \geq t)$$

- Here, s represents the time you've already spent waiting.
- The above states that after you've waited s time units, the probability you'll need to wait an **additional** t time units is equal to the probability you'd have to wait t time units without having waited those s time units in the first place.
- Example: If train arrival is guided by a memoryless random variable, the fact that you've waited 15 minutes doesn't obligate the train to arrive any faster!

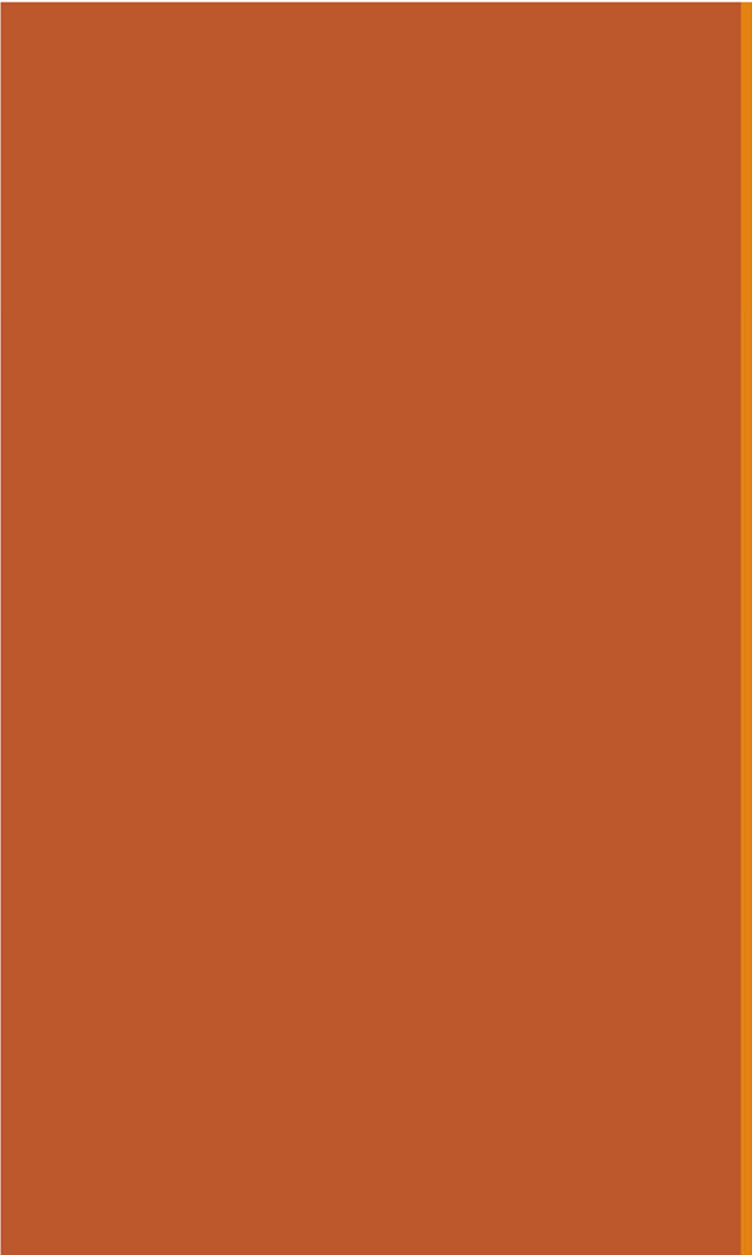
Memorylessness: Hurry Up and Wait

A continuous probability distribution is said to be **memoryless** if a random variable X on that probability distribution satisfies the following for all $s, t \geq 0$:

$$P(X \geq s + t \mid X \geq s) = P(X \geq t)$$

Using the definition of conditional probability, we can show that our Exponential distribution possesses the memoryless property. Just let $X \sim \text{Exp}(\lambda)$ and trust the math:

$$P(X \geq s + t \mid X \geq s) = \frac{P(X \geq s + t)}{P(X \geq s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} = P(X \geq t)$$



Exercises

Earthquakes

Major earthquakes (magnitude 8.0+) occur independently on average once every 500 years.*

What is the probability of **zero major earthquakes next year**?

Strategy 1: Exponential RV

Define events/RVs & state goal

T : when first earthquake happens

$T \sim \text{Exp}(\lambda = 0.002)$

Want: $P(T > 1) = 1 - F(1)$

Solve

$$P(T > 1) = 1 - (1 - e^{-\lambda \cdot 1}) = e^{-\lambda} \\ = e^{-0.002} \approx 0.998$$

*In California, according to historical data from USGS, 2015

Earthquakes

$$Y \sim \text{Poi}(\lambda) \quad p(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Major earthquakes (magnitude 8.0+) occur independently on average once every 500 years.*

What is the probability of **zero major earthquakes next year**?

Strategy 1: Exponential RV

Define events/RVs & state goal

T : when first earthquake happens

$$T \sim \text{Exp}(\lambda = 0.002)$$

Want: $P(T > 1) = 1 - F(1)$

Solve

$$P(T > 1) = 1 - (1 - e^{-\lambda \cdot 1}) = e^{-\lambda}$$

Strategy 2: Poisson RV

Define events/RVs & state goal

N : # earthquakes next year

$$N \sim \text{Poi}(\lambda = 0.002)$$

Want: $P(N = 0)$

$$\lambda: \frac{\text{earthquakes}}{\text{year}}$$

Solve

$$P(N = 0) = \frac{\lambda^0 e^{-\lambda}}{0!} = e^{-\lambda} \approx 0.998$$

*In California, according to historical data from USGS, 2015

Replacing your laptop

$$X \sim \text{Exp}(\lambda) \quad \begin{aligned} E[X] &= 1/\lambda \\ F(x) &= 1 - e^{-\lambda x} \end{aligned}$$

Let X = # hours of use until your laptop dies.

- X is distributed as an Exponential RV, where
- On average, laptops die after 5000 hours of use.
- You use your laptop 5 hours a day.

What is $P(\text{your laptop lasts 4 years})$?

Define

X : # hours until
laptop death
 $X \sim \text{Exp}(\lambda = 1/5000)$

Want: $P(X > 5 \cdot 365 \cdot 4)$

Solve

$$\begin{aligned} P(X > 7300) &= 1 - F(7300) \\ &= 1 - (1 - e^{-7300/5000}) = e^{-1.46} \approx 0.2322 \end{aligned}$$

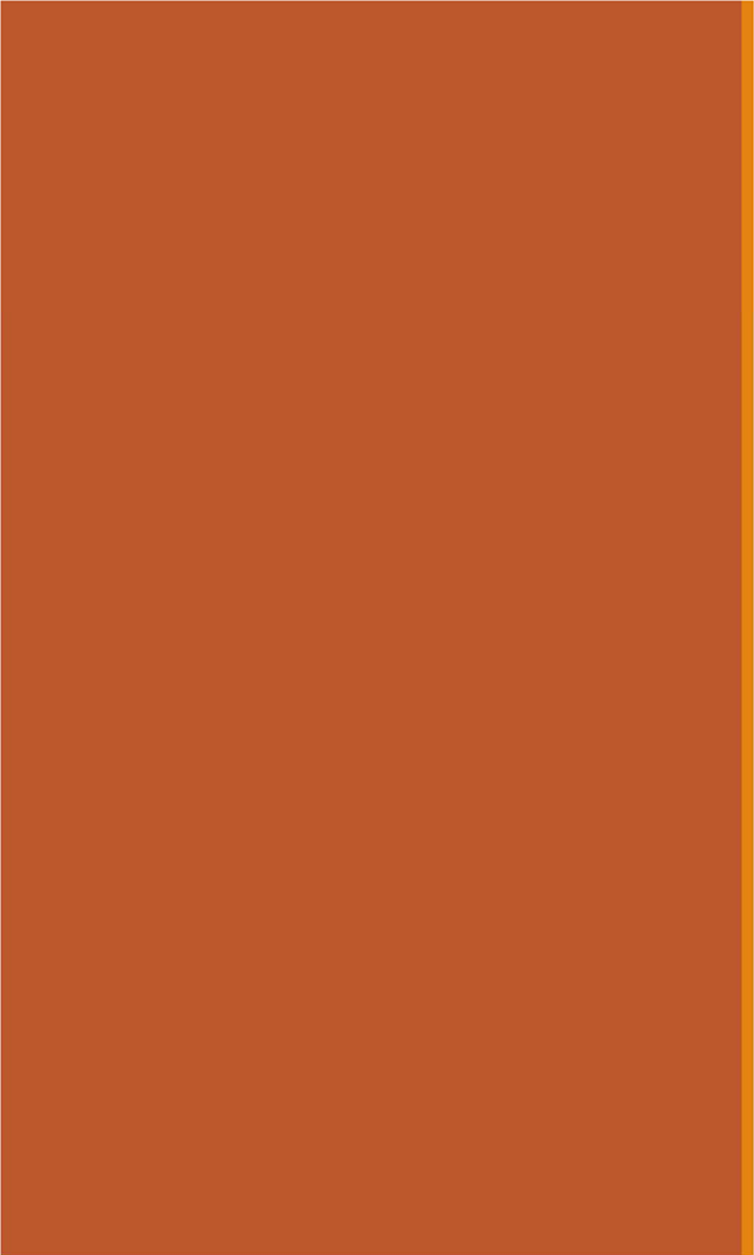
Better plan ahead if you're co-termining!

- 5-year plan:

$$P(X > 9125) = e^{-1.825} \approx 0.1612$$

- 6-year plan:

$$P(X > 10950) = e^{-2.19} \approx 0.1119$$



Extra

Expectation of the Exponential

$$X \sim \text{Exp}(\lambda) \quad f(x) = \lambda e^{-\lambda x} \quad \text{if } x \geq 0$$

$$X \sim \text{Exp}(\lambda)$$

Expectation

$$E[X] = \frac{1}{\lambda}$$

Proof:

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x \lambda e^{-\lambda x} dx$$

$$= -x e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

$$= -x e^{-\lambda x} \Big|_0^{\infty} - \frac{1}{\lambda} e^{-\lambda x} \Big|_0^{\infty}$$

$$= [0 - 0] + \left[0 - \left(-\frac{1}{\lambda} \right) \right]$$

$$= \frac{1}{\lambda}$$

Integration by parts

$$\int x \lambda e^{-\lambda x} dx = \int u \cdot dv$$

$$\begin{aligned} u &= x & dv &= \lambda e^{-\lambda x} dx \\ du &= dx & v &= -e^{-\lambda x} \end{aligned}$$

$$\int u \cdot dv = u \cdot v - \int v \cdot du$$

$$-x e^{-\lambda x} - \int -e^{-\lambda x} dx$$